



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 7 • STANDARD 6.NOS.7

Ordered Pairs in All Four Quadrants

Lesson 7-8 • Teacher Copy

Name: _____

Date: _____

Class: _____

My Notes

Level 3 Enrichment



TODAY'S OBJECTIVES

Content Objective: I can plot ordered pairs in all four quadrants of the coordinate plane.

Language Objective: I can explain my work using the words quadrant, negative coordinate, reflection, and axis.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Quadrant

Negative coordinate

Reflection

Axis

Integer



Tap any word to see its meaning and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

One of the four regions of the coordinate plane, numbered I to IV, is a

2

A coordinate less than zero, which places a point left of or below the origin, is a

3

A flip of a point over an axis is a

4

One of the two number lines, the x-axis or y-axis, that form the grid is an

5

A whole number or its opposite is an

Write About the Math

How do four-quadrant coordinates help a city map locations in every direction from the center?

¿Cómo ayudan las coordenadas de cuatro cuadrantes a una ciudad a ubicar lugares en todas las direcciones desde el centro?

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **Quadrant**
Cuadrante

☐ **Negative coordinate**
Coordenada negativa

☐ **Reflection**
Reflexión

☐ **Axis**
Eje

☐ **Integer**
Número entero

Model: *The signs of the coordinates tell me the quadrant because ____.* — Student matches sign combinations to quadrants (e.g., (+,+) is I, (-,+) is II).

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The library is in Quadrant ____ because its x is ____ and its y is ____.** The park is in Quadrant ____ because ____.
- **My answer is ____ because ____.**
Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
Mi afirmación es ____.
- **My evidence is ____, so ____.**
Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Quadrant
2. **Notes 2:** Negative coordinate
3. **Notes 3:** Reflection
4. **Notes 4:** Axis
5. **Notes 5:** Integer
6. **Watch for this mistake:** *A common mistake in Ordered Pairs in All Four Quadrants is matching each sign to the wrong axis — for example, seeing $(-2, 5)$ and thinking "the x is negative so I move DOWN, and the y is positive so I move RIGHT," which lands the point in Quadrant IV. But the x -coordinate always controls left/right, never up/down: negative x means LEFT (not down), and positive y means UP (not right). So $(-2, 5)$ is actually left 2, up 5 — Quadrant II. Before you submit, check that you used the x -coordinate for left/right and the y -coordinate for up/down, and never swapped them.*
7. **Try It 1:** D. Quadrant III — *A negative x means left and a negative y means down, so $(-5, -2)$ is in Quadrant III, where both coordinates are negative.*
8. **Try It 2:** A. $(-, +)$ — *Quadrant II is up and to the left of the origin, so x is negative and y is positive: $(-, +)$.*